

Use of Variable in Conf File

To Begin with the Lab

Summary of the Lab

This lab demonstrates how Terraform variables make infrastructure configurations more flexible and easier to manage. Instead of hardcoding values such as AWS credentials and region settings, variables are used as placeholders and their values are supplied when Terraform runs. The lab covers three ways of providing variable values: interactive prompts, command-line arguments, and a terraform.tfvars file. This approach improves reusability, organization, and security of Terraform configurations.

- Steps
 - Create a new project folder for the Terraform variable lab.
 - Create create-instance.tf and define an EC2 instance resource.
 - Create provider.tf and configure the AWS provider using variables for credentials and region.
 - Create variables.tf and define access_key, secret_key, and region variables.
 - Set a default value for the region variable.
 - Pull the latest files to the Terraform machine and navigate to the project directory.
- Run:

```
terraform init
```

- Execute:

```
terraform plan
```

- Terraform detects that access_key and secret_key have no values.
- It prompts for them interactively.
- Enter the required values when asked.
- Terraform uses the default region value (us-east-1).
- Verify Terraform Plan
- If invalid credentials are entered, Terraform returns an authentication error.
- If valid credentials are entered, Terraform successfully displays the execution plan.
- Run Terraform Plan Using Command-Line Variables
- Instead of entering values interactively, provide them directly:

```
terraform plan \ -var="access_key=YOUR_ACCESS_KEY" \ -  
var="secret_key=YOUR_SECRET_KEY"
```

- Terraform reads the values from the command line.
- No prompts appear.
- Create a Variable Values File
- Create a file named:

- terraform.tfvars
- Add:
 - access_key = "YOUR_ACCESS_KEY"
 - secret_key = "YOUR_SECRET_KEY"
- Run:

```
terraform plan
```

- To verify Terraform automatically reads values from the variable file.
- Review the generated execution plan and project files.